



COURSE SLIDES

Generative AI for Sustainability Reporting

Course Code: C1311

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Course Trainer Profiles

1

Ng Herk Low

Sustainability professional and trainer with leadership experience in strategy, governance, social impact and stakeholder-focused sustainability reporting.

2

Lynn Foo

Sustainability and ESG practitioner with experience in carbon measurement, reporting, environmental management, circular economy and organisational governance.

Let's Know Each Other

1 Your name and organisation / role.

2 Your experience with this subject (if any).

3 What you want to be able to do after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

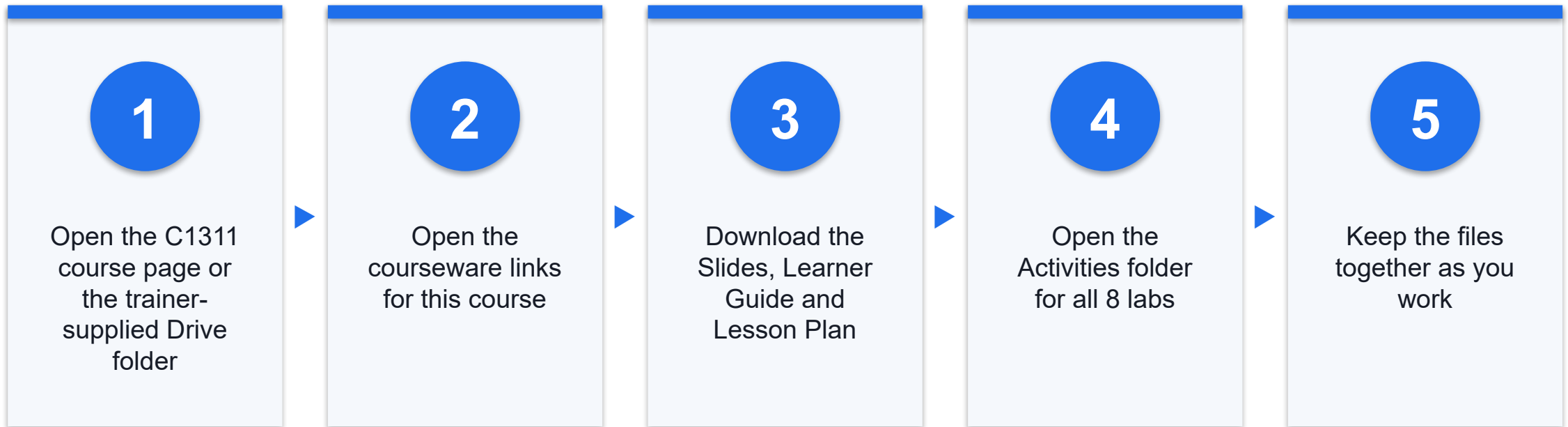
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Access Course Materials



Course page: <https://www.tertiarycourses.com.sg/generative-ai-for-sustainability-reporting.html>

Lesson Plan — 2 Days, 15 instructional hours

Course sequence

- Day 1 — Ground the AI workflow and turn ESG evidence into trustworthy metrics
 - Topic 1: Getting Started with Generative AI for Sustainability Reporting (Labs 1–2)
 - Topic 2: AI for ESG Data Collection and Analysis (Labs 3–5)

Day 2

- Day 2 — Turn verified evidence into framework-mapped, review-ready reporting
 - Topic 3: AI for Drafting and Structuring Reports (Lab 6)
 - Topic 4: AI for Compliance, Frameworks and Continuous Reporting (Labs 7–8)
- Daily timing
 - 9:30 am – 6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)

Learning Outcomes

1

Ground & Govern

Establish a responsible, evidence-controlled generative AI workflow and distinguish the purposes of GRI, SASB, TCFD and ISSB reporting guidance.

2

Structure & Analyse

Structure ESG source data, calculate and interpret selected emissions and intensity metrics, and perform transparent materiality and gap analysis.

3

Draft & Validate

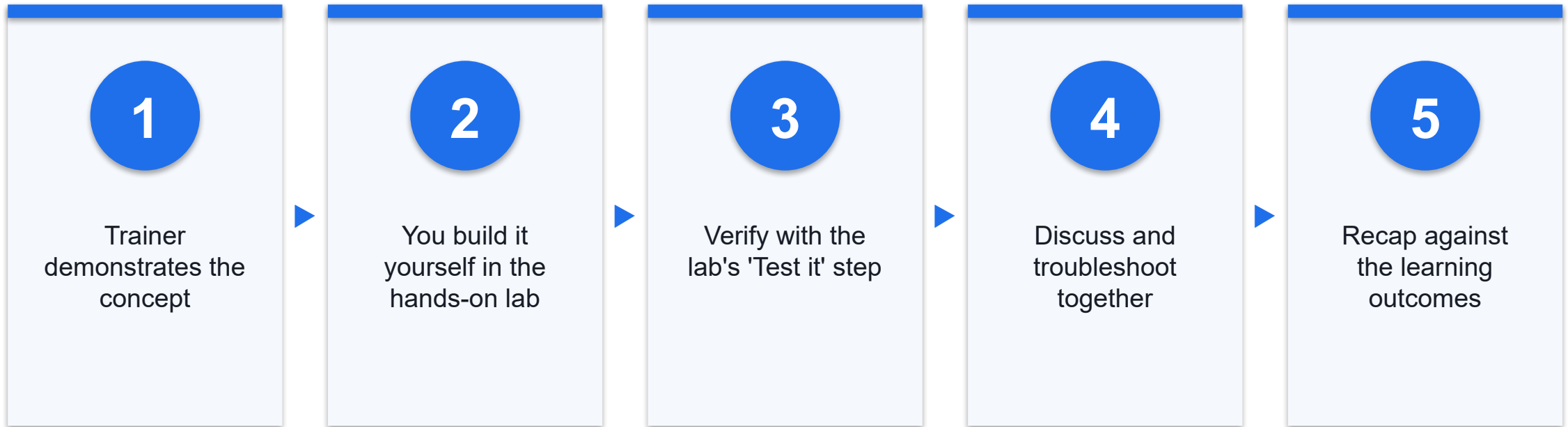
Draft readable, evidence-grounded sustainability disclosures and executive summaries, then validate claims, consistency and limitations.

4

Map & Operationalise

Map report content to applicable frameworks, maintain an assurance-ready audit trail, and design a governed recurring reporting workflow.

How You'll Learn





CORE CONCEPTS

The Evidence-Controlled AI Reporting System

Six Ideas That Keep Reporting Credible

1

Reporting basis

Audience, period, boundary, materiality lens, framework and applicability.

2

Evidence first

Stable source IDs, raw values, owners, status and immutable originals.

3

Transparent calculations

Activity data, factor, formula, conversion, result and reviewer.

4

Bounded AI tasks

Transform, compare, draft or critique supplied evidence—one controlled task at a time.

5

Human judgement

People decide materiality, applicability, estimates, balance and publication.

6

Reproducible lineage

Source → transformation → calculation → claim → review → approval.

The Eight-Stage Reporting Workflow

1

1 · Scope

Define reporting basis, boundary, audience and applicability.

2

2 · Register

Inventory sources, owners, periods, units and evidence status.

3

3 · Structure

Normalise data while retaining raw values and provenance.

4

4 · Calculate

Apply documented methods, factors, conversions and checks.

5

5 · Prioritise

Assess materiality and framework gaps with explicit anchors.

6

6 · Draft

Generate source-led sections and claim ledgers.

The Eight-Stage Reporting Workflow (cont.)

7

7 · Review

Reconcile claims, mappings, controls and approvals.

8

8 · Publish

Archive the evidence pack and improve the next cycle.

OPERATING PRINCIPLE

AI accelerates reporting work. It does not own the evidence, judgement or conclusion.

Credibility comes from traceable sources, transparent calculations, current requirements and accountable human review.

What You Will Build

Controlled evidence

- AI use charter and prompt log
- Framework and applicability register
- Structured ESG dataset and data dictionary

Verified reporting

- Emissions and intensity workpaper
- Materiality and gap matrix
- Disclosure draft and claim ledger

Repeatable governance

- Framework crosswalk
- Assurance evidence index
- Recurring reporting workflow board

The Review Loop Used in Every Lab

1

Ground the task in synthetic evidence and an explicit reporting boundary.

2

Ask for a structured transformation, calculation check, draft or critique.

3

Compare output with original sources, formulas and current official requirements.

4

Record unresolved items and make the human decision to approve, revise or stop.

5

Carry the versioned artifact and evidence trail into the next lab.

Keep the Materiality Lenses Separate

1

Impact lens

Which impacts on the economy, environment and people are most significant?

2

Investor lens

Which sustainability-related risks and opportunities could affect the entity's prospects?

3

Shared evidence

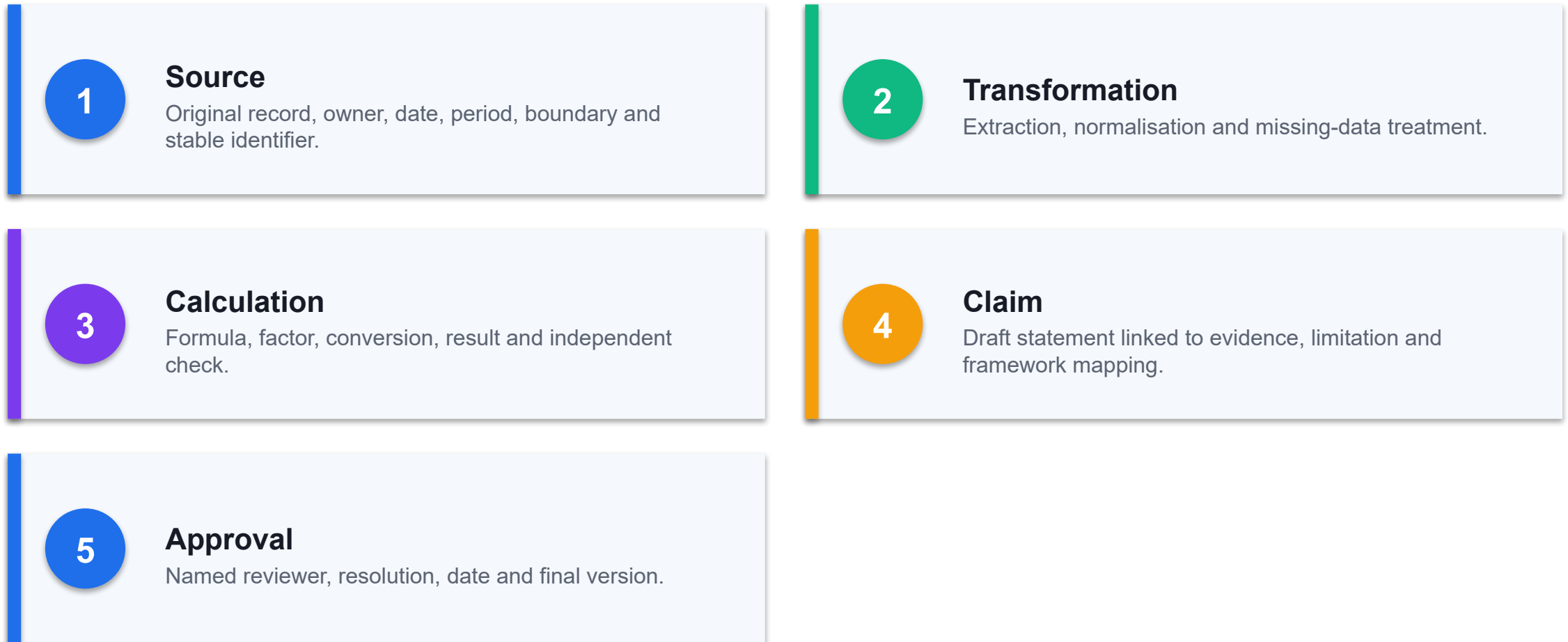
One source may inform both lenses, but each conclusion needs its own rationale.

4

Documented judgement

Thresholds, stakeholders, owners and decisions remain visible.

The Reporting Evidence Chain



TOPIC 01

Getting Started with Generative AI for Sustainability Reporting

ESG and sustainability reporting · AI-assisted work · reporting frameworks · prompt design · responsible and transparent use

Key Concepts — Getting Started with Generative AI for Sustainability Reporting

1

Reporting purpose

Connect material impacts, risks and opportunities to decision-useful, traceable disclosures.

2

Framework lenses

Use GRI for significant impacts and ISSB/SASB for investor-focused risks and opportunities; retain TCFD as a useful climate bridge.

3

Grounded generation

Ask an AI assistant to transform supplied evidence, never to manufacture missing evidence.

4

Human accountability

A named preparer reviews scope, calculations, claims, privacy, rights and applicability before use.

5

Prompt contract

State goal, reporting context, constraints, sources, output schema and review tests.

6

Transparent use

Record model, date, inputs, revisions, source links and final human decision.

ESG, Sustainability Reporting and Generative AI

WHAT

- Sustainability reporting explains an organisation's material impacts, risks, opportunities, governance, actions and performance. Generative AI is a drafting and transformation aid within that process; it is not a source of corporate evidence.

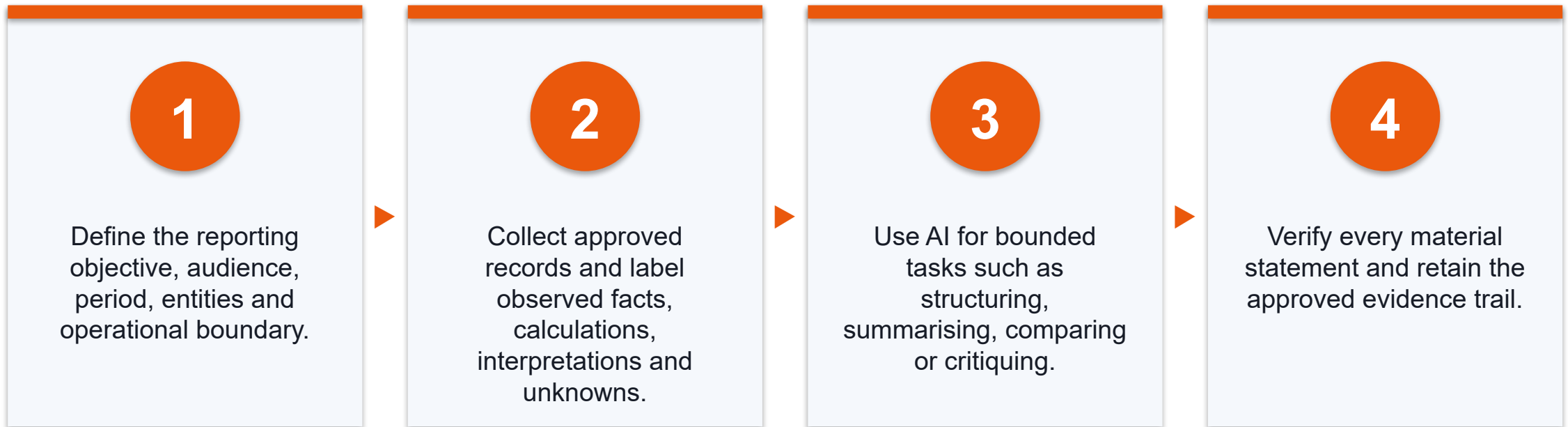
WHY IT MATTERS

- Readers rely on reported information for decisions. A fluent AI response can hide unsupported claims, boundary errors or omitted uncertainty unless the reporting team keeps evidence and judgement separate from language generation.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — ESG, Sustainability Reporting and Generative AI



In Practice — ESG, Sustainability Reporting and Generative AI

WORKED EXAMPLE

- HarbourLight Foods has electricity invoices, diesel records and a prior-year narrative.
- The assistant organises those records into a disclosure outline and labels missing data.
- The preparer calculates the metrics, confirms the boundary and approves the final wording.

DECISION GUIDE

- **Use when**
 - The task has a clear owner, approved evidence and a review checklist.
 - The output can be traced back to a source record or transparent calculation.
- **Avoid when**
 - The assistant would receive confidential or personal data outside approved controls.
 - The task asks the model to invent a figure, legal conclusion or assurance opinion.

Quality Lens — ESG, Sustainability Reporting and Generative AI

Failure signal

- The draft contains precise claims but no source identifiers or boundary statement.

Repair move

- Add an evidence table, explicit unknown labels and a named human approval gate.

Quality evidence

- Each material claim links to a record, calculation or documented judgement.

Set Up AI Tools for Reporting

WHAT

- An AI reporting workspace combines an approved assistant, a synthetic or authorised source pack, a structured working folder and a versioned log of prompts, outputs and human decisions.

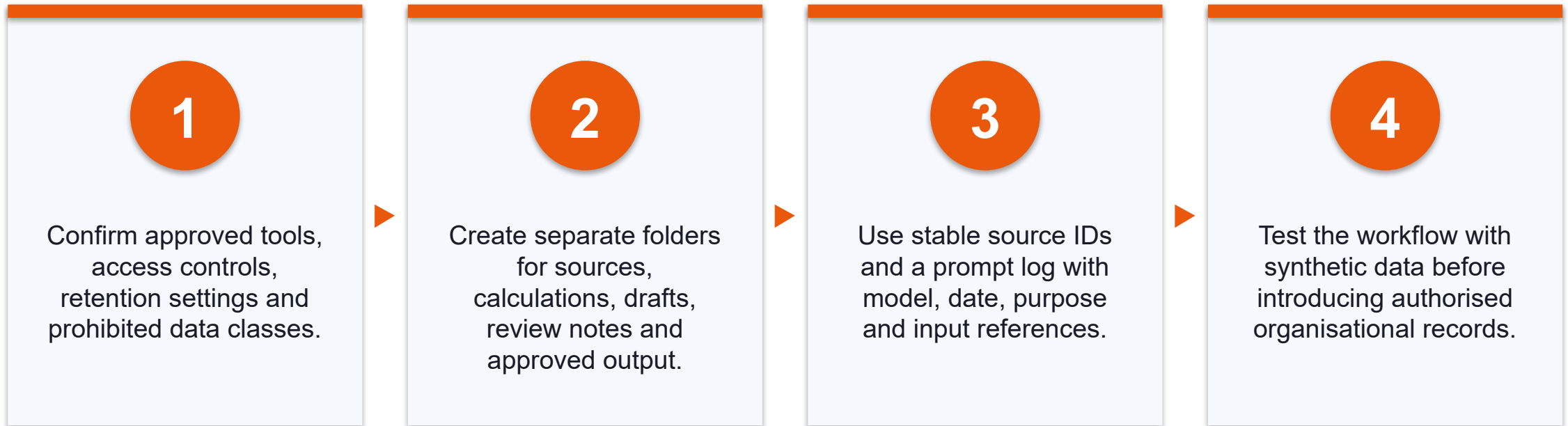
WHY IT MATTERS

- Setup choices determine what data is exposed, whether work can be reproduced and whether later reviewers can distinguish raw evidence from AI-generated working text.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Set Up AI Tools for Reporting



In Practice — Set Up AI Tools for Reporting

WORKED EXAMPLE

- The team creates HLF-2025/source, calculation, draft and review folders.
- Invoice INV-E-012 and factor EF-SG-2025-TRAINING remain referenced by stable IDs.
- Prompt P-001 uses only synthetic extracts and the reviewer records APPROVE, REVISE or STOP.

DECISION GUIDE

- **Use when**
 - The organisation has approved the tool and defined its data-handling rules.
 - A reviewer can reproduce the task from saved source IDs and prompt instructions.
- **Avoid when**
 - Consumer accounts are used for restricted records without organisational approval.
 - Source files and generated drafts are mixed in one unversioned folder.

Quality Lens — Set Up AI Tools for Reporting

Failure signal

- No one can tell which input or model produced a sentence.

Repair move

- Introduce stable IDs, version names, a prompt log and separate source and draft areas.

Quality evidence

- A second reviewer can reproduce the draft lineage without guessing.

GRI, SASB, TCFD and ISSB at a Glance

WHAT

- GRI focuses on an organisation's most significant impacts on the economy, environment and people. ISSB Standards establish an investor-focused baseline; SASB supplies industry-based topics and metrics. TCFD's governance, strategy, risk management, and metrics-and-targets architecture is incorporated into IFRS S2.

WHY IT MATTERS

- Framework names are not interchangeable. Selecting a disclosure only because an AI response mentioned a familiar acronym can create materiality, audience and completeness errors.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — GRI, SASB, TCFD and ISSB at a Glance



In Practice — GRI, SASB, TCFD and ISSB at a Glance

WORKED EXAMPLE

- Water stress may be significant as an impact under GRI even when it is not financially material.
- A climate transition risk may be material to investors under IFRS S1 and IFRS S2.
- A food-sector SASB metric can add industry specificity while the TCFD architecture supports transition to IFRS S2.

DECISION GUIDE

- **Use when**
 - Building a framework register or explaining why a disclosure belongs in the report.
 - Reconciling impact-focused and investor-focused information without merging their tests.
- **Avoid when**
 - Declaring broad compliance from a keyword match or incomplete checklist.
 - Treating TCFD as an additional current layer where IFRS S2 already covers it, unless a rule still requires it.

Quality Lens — GRI, SASB, TCFD and ISSB at a Glance

Failure signal

- The framework column contains several acronyms but no requirement IDs or rationale.

Repair move

- Record audience, lens, official source, requirement identifier and applicability decision.

Quality evidence

- Every mapping explains both why the item is relevant and what evidence supports it.

Effective Prompting and Responsible, Transparent AI Use

WHAT

- A reporting prompt is a controlled work instruction containing a goal, context, constraints, source boundaries, output schema and review criteria. Transparent use records the assistant's role and the human checks that converted working text into approved content.

WHY IT MATTERS

- Generic prompts reward plausible prose. A structured prompt and adversarial review make missing evidence, inconsistent units, unsupported causal language and invented requirements easier to detect.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Effective Prompting and Responsible, Transparent AI Use



In Practice — Effective Prompting and Responsible, Transparent AI Use

WORKED EXAMPLE

- Goal: draft a 150-word energy disclosure for FY2025 using sources HLF-E01 to HLF-E06.
- Constraint: preserve units, label calculated values and write MISSING where evidence is absent.
- Review: return a claim ledger and flag wording that implies causation, certainty or compliance.

DECISION GUIDE

- **Use when**
 - Drafting, restructuring or reviewing content from a bounded evidence pack.
 - The output format exposes uncertainty and supports line-by-line verification.
- **Avoid when**
 - The prompt asks for a current legal or standards conclusion without checking the official source.
 - The model is asked to conceal its role or remove known caveats to sound more confident.

Quality Lens — Effective Prompting and Responsible, Transparent AI Use

Failure signal

- The answer is polished but fills evidence gaps silently.

Repair move

- Require MISSING labels, a claim ledger and an independent human source check.

Quality evidence

- Uncertainty remains visible and all accepted claims survive source verification.

Hands-On Labs — Getting Started with Generative AI for Sustainability Reporting

Lab 1

- Set the Reporting Boundary and AI Control Contract

Lab 2

- Build the Framework and Materiality-Lens Register

Set the Reporting Boundary and AI Control Contract

LAB 1 · 40
MIN

You will set up the synthetic HarbourLight Foods reporting workspace, define the reporting entity, period, audience and evidence boundary, and write an AI-use charter. You will then run one bounded transformation task and record the sources, output and human decision.

You'll build

A reporting-basis.md, ai-use-charter.md and prompt-log.csv that define scope, permitted data, prompt controls, review gates and a reproducible first AI-assisted task.

1 Define the basis

2 Set AI controls

3 Write the prompt contract

4 Test and log

Tools: Text editor · approved AI assistant · labs/assets/harbourlight-company-brief.md · source-pack-index.md

Set the Reporting Boundary and AI Control Contract

Test it

Open the three control files. reporting-basis.md must state entity, FY2025 period, boundary, audiences, owners and unresolved questions. ai-use-charter.md must contain all five headings and at least fifteen specific rules. prompt-log.csv must contain P-001 with source HLF-BRIEF-01, output filename, reviewer, decision and reason. P-001-output.md must have no unsupported row marked SUPPORTED.

Build the Framework and Materiality-Lens Register

LAB 2 · 45
MIN

You will inspect the supplied official-source links, complete a framework register for GRI, ISSB, SASB and TCFD, and add Singapore applicability checks. An AI assistant may compare the supplied summaries, but you will verify every retained statement against the official page.

You'll build

A framework-and-applicability-register.csv plus materiality-lenses.md showing purpose, audience, lens, requirement source, effective-date check, applicability owner and unresolved questions.

1 Open official sources

2 Classify the lenses

3 Map reporting needs

4 Critique and verify

Tools: Browser · spreadsheet · approved AI assistant · framework-starter-register.csv

Build the Framework and Materiality-Lens Register

Test it

The register must contain all nine required framework or jurisdiction rows, official URL, and either an actual checked date or PENDING CHECK with a named owner. Each row must have audience, lens, current-status note and owner. It must contain five HarbourLight reporting-need rows and no unsupported applicability claim marked CURRENT. materiality-lenses.md must keep the two lenses separate, explain shared evidence correctly and identify the owner of final applicability decisions. prompt-log.csv must contain P-002 and the register must record a reviewer result and reason for every AI-proposed change.

Recap — Getting Started with Generative AI for Sustainability Reporting

1 You can now: LO1: establish a responsible, evidence-controlled generative AI reporting workflow

2 You can now: LO1: distinguish GRI, SASB, TCFD and ISSB purposes and record current applicability

TOPIC 02

AI for ESG Data Collection and Analysis

Source registers · data dictionaries · metric calculations · emissions boundaries · materiality · gap analysis · charts and summaries

Key Concepts — AI for ESG Data Collection and Analysis

1

Evidence register

Track source owner, period, boundary, unit, status and location before analysis.

2

Data dictionary

Define each field, unit, allowed value, calculation and missing-data treatment.

3

Calculation lineage

Preserve activity data, factor source, formula, unit conversion and reviewer.

4

Materiality lenses

Assess significant impacts separately from investor-focused risks and opportunities.

5

Gap status

Distinguish complete, partial, missing, not applicable and pending verification.

6

Visual integrity

Use consistent denominators, labelled units, honest scales and visible limitations.

Gather and Structure ESG Data with AI

WHAT

- ESG data collection turns heterogeneous records into a controlled dataset with stable identifiers, defined fields, consistent units, reporting boundaries and visible evidence status.

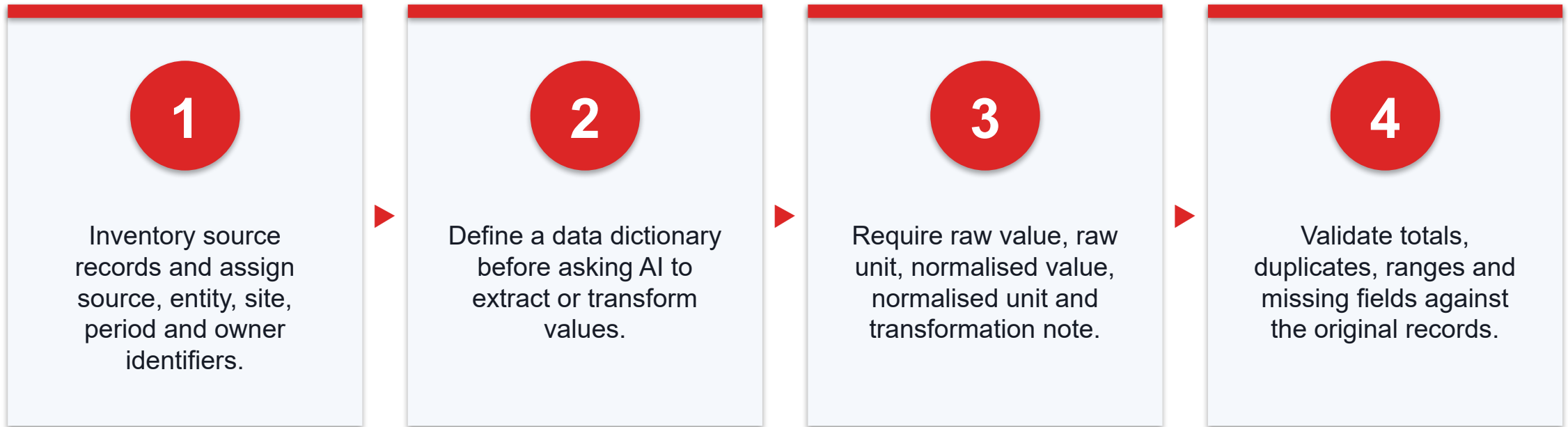
WHY IT MATTERS

- AI can extract or normalise repeated fields quickly, but unreviewed extraction may change units, merge entities, duplicate records or convert blanks into invented values.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Gather and Structure ESG Data with AI



In Practice — Gather and Structure ESG Data with AI

WORKED EXAMPLE

- Three sites provide electricity in kWh, MWh and invoice line-item extracts.
- The target schema normalises energy to kWh while retaining raw values and source pages.
- A pivot check compares extracted totals with invoice totals and flags one duplicate month.

DECISION GUIDE

- **Use when**
 - The source set is bounded and the extraction schema is explicit.
 - A human can sample and reconcile extracted values to original records.
- **Avoid when**
 - Scanned records are unreadable or the model cannot preserve page-level provenance.
 - The workflow overwrites raw values or discards missing-data reasons.

Quality Lens — Gather and Structure ESG Data with AI

Failure signal

- Normalised values exist without raw values or transformation notes.

Repair move

- Restore immutable raw columns, units, source IDs and reconciliation checks.

Quality evidence

- A reviewer can trace every reported number to the original record and conversion.

Calculate and Interpret Metrics and Emissions

WHAT

- A sustainability metric combines a precisely defined numerator, denominator, boundary, period and unit. A basic greenhouse-gas calculation multiplies activity data by an appropriate emission factor and converts units consistently.

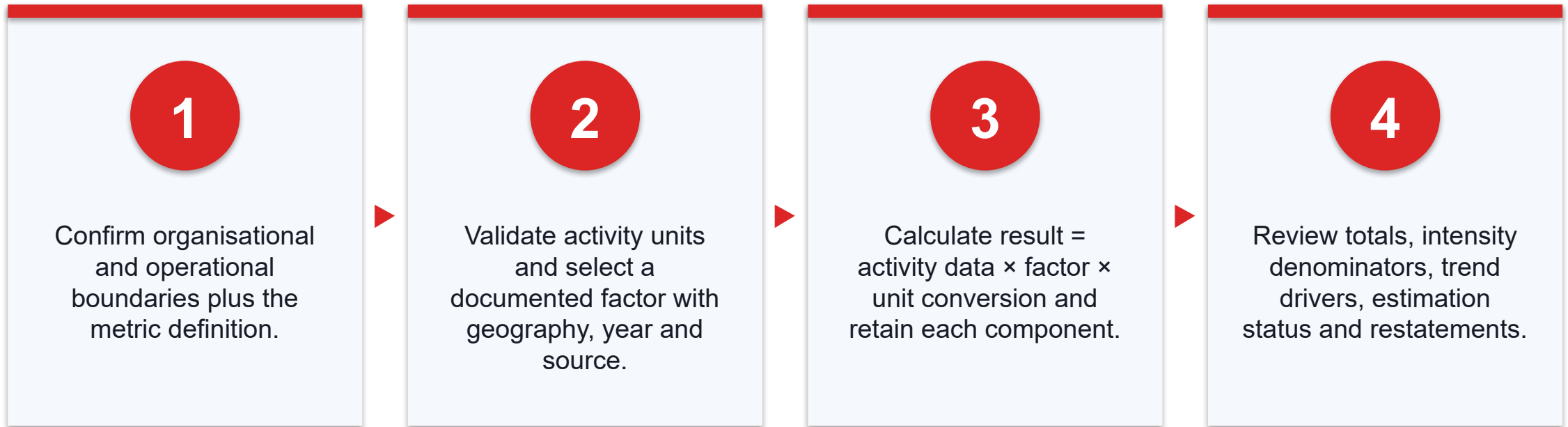
WHY IT MATTERS

- Seemingly small choices—scope, factor year, consolidation boundary, denominator or rounding—can materially change a trend and make year-on-year comparison misleading.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Calculate and Interpret Metrics and Emissions



In Practice — Calculate and Interpret Metrics and Emissions

WORKED EXAMPLE

- $520,000 \text{ kWh} \times 0.408 \text{ kg CO}_2\text{e/kWh} = 212,160 \text{ kg CO}_2\text{e}$, or 212.16 tCO₂e.
- Revenue intensity = total tCO₂e ÷ S\$ million revenue, using the same reporting period.
- The narrative states that the electricity factor is a synthetic training value, not an official inventory factor.

DECISION GUIDE

- **Use when**
 - Inputs, factors, conversions and boundaries are documented and reviewed.
 - Comparisons use consistent periods and explain any recalculation or estimation.
- **Avoid when**
 - A model supplies an emission factor without an authoritative, current source.
 - An intensity improvement is presented without also checking the absolute metric and denominator change.

Quality Lens — Calculate and Interpret Metrics and Emissions

Failure signal

- A result has no unit, factor version, boundary or calculation trail.

Repair move

- Rebuild the calculation table from activity data through conversion to final metric.

Quality evidence

- The figure recalculates exactly and its comparison basis is explicit.

Conduct Materiality and Gap Analysis

WHAT

- Materiality analysis prioritises significant impacts and/or sustainability-related risks and opportunities using the stated reporting lens. Gap analysis compares required or chosen disclosures with available evidence.

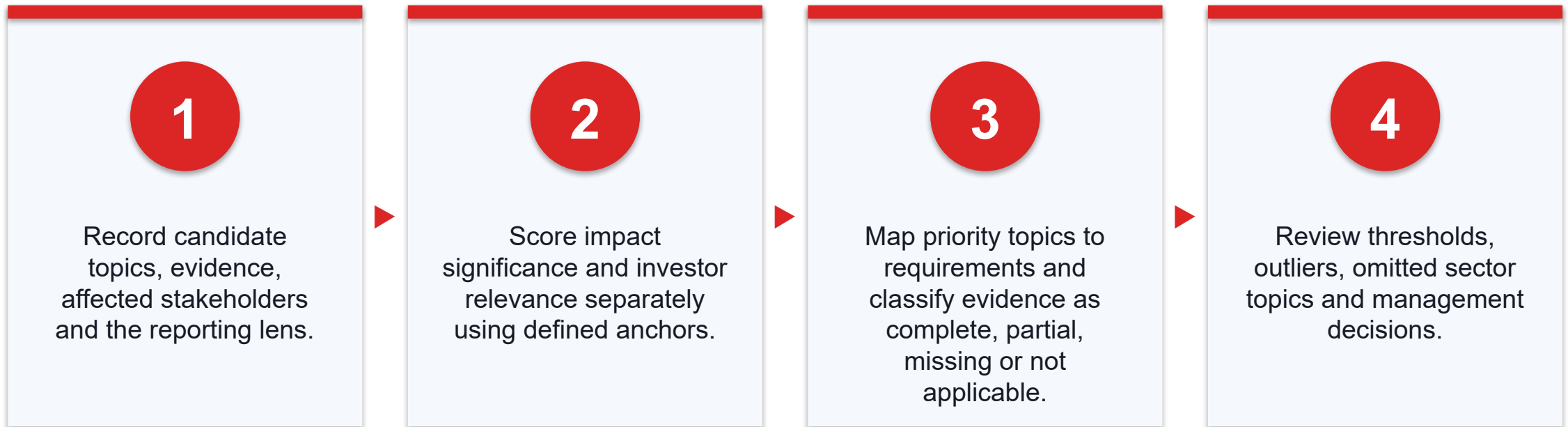
WHY IT MATTERS

- AI can cluster evidence and surface candidate topics, but it cannot replace stakeholder engagement, management judgement, legal applicability decisions or the documented basis for prioritisation.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Conduct Materiality and Gap Analysis



In Practice — Conduct Materiality and Gap Analysis

WORKED EXAMPLE

- Packaging waste scores high for environmental impact while energy price volatility scores high for financial effect.
- The framework register shows complete energy data but only partial supplier-screening evidence.
- The decision log records the threshold, reviewer and reason for each included or deferred item.

DECISION GUIDE

- **Use when**
 - Scoring anchors are written before scoring and evidence remains attached to each judgement.
 - The output supports a decision log rather than claiming to automate materiality.
- **Avoid when**
 - One combined score hides the difference between impact and financial materiality.
 - Stakeholder views or sector guidance are generated instead of collected and verified.

Quality Lens — Conduct Materiality and Gap Analysis

Failure signal

- Every topic is labelled material with identical reasoning.

Repair move

- Separate lenses, define scoring anchors and allow evidence-based non-priority decisions.

Quality evidence

- A reviewer can reproduce the ranking and challenge each judgement.

Visualise and Summarise ESG Data

WHAT

- A reporting visual encodes a defined metric across time, entity or category so a reader can compare values without losing units, denominators, boundaries or uncertainty.

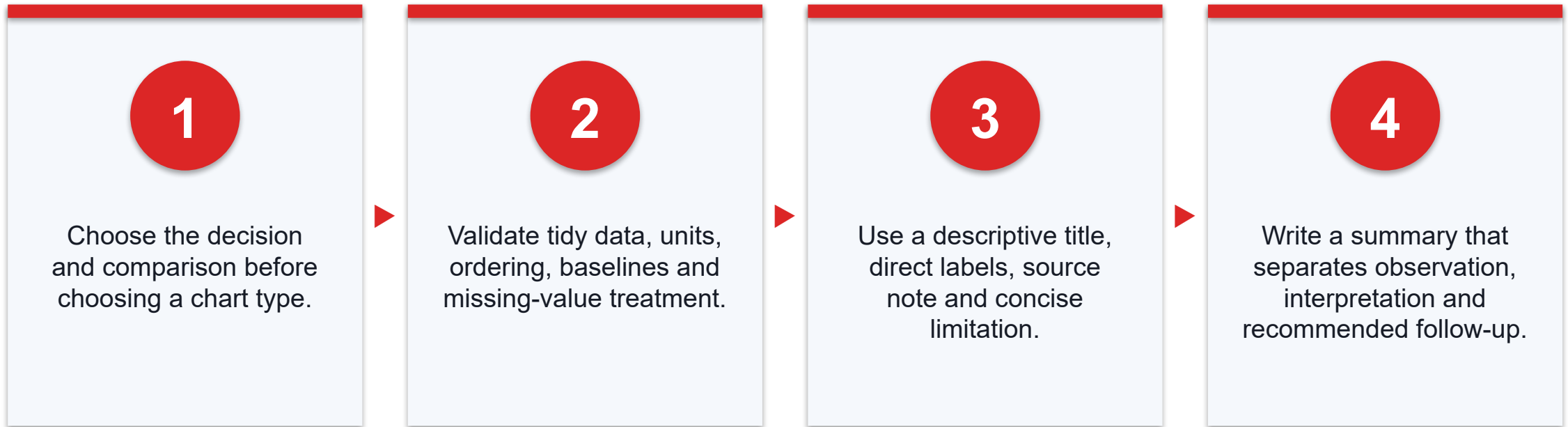
WHY IT MATTERS

- Charts compress complex data, which also makes them capable of hiding denominator shifts, missing periods, inconsistent scales and estimation changes.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Visualise and Summarise ESG Data



In Practice — Visualise and Summarise ESG Data

WORKED EXAMPLE

- A two-year bar chart shows absolute Scope 1 and Scope 2 tCO₂e by year.
- A small companion table shows revenue and tCO₂e per S\$ million to expose denominator effects.
- The summary notes the observed change and asks for operations evidence before assigning a cause.

DECISION GUIDE

- **Use when**
 - The visual materially improves comparison and all plotted fields share a coherent grain.
 - Readers can see units, time period, boundary, source and limitations.
- **Avoid when**
 - A decorative chart replaces a short table or implies precision unsupported by the data.
 - A causal explanation is generated from correlation or two aggregated points.

Quality Lens — Visualise and Summarise ESG Data

Failure signal

- The chart has a clever title but no unit, boundary or source note.

Repair move

- Restore neutral labelling, units, data notes and a separate interpretation sentence.

Quality evidence

- The visual and source table reconcile exactly and support the same conclusion.

Hands-On Labs — AI for ESG Data Collection and Analysis

Lab 3

- Create the ESG Evidence Register and Data Dictionary

Lab 4

- Calculate Scope 1, Scope 2 and Intensity Metrics

Lab 5

- Run Materiality, Gap and Visual Analysis

Create the ESG Evidence Register and Data Dictionary

LAB 3 · 50
MIN

You will inspect HarbourLight's synthetic activity records, define a data dictionary and create a structured ESG dataset. An AI assistant may propose field classifications and normalisation rules, but you will reconcile every retained value to its source row.

You'll build

A data-dictionary.csv, esg-evidence-register.csv and structured-esg-data.csv containing raw and normalised values, source IDs, owners, periods, boundaries and review status.

1 Inventory sources

2 Define the schema

3 Normalise carefully

4 Reconcile and log

Tools: Spreadsheet · approved AI assistant · esg-activity-data.csv · esg-evidence-register-starter.csv

Create the ESG Evidence Register and Data Dictionary

Test it

data-dictionary.csv must define all required fields and their controls. Every structured row must retain Row_ID, Controlled_Metric, raw value, raw unit, source ID, source row and unique Transformation_ID. All MWh rows must convert exactly to kWh, no other value may change without a rule, and every source lookup must resolve. The control log must contain C01–C04 with reviewer, result and resolution or named open owner. prompt-log.csv must contain completed P-003 review fields.

Calculate Scope 1, Scope 2 and Intensity Metrics

LAB 4 · 55
MIN

You will join verified activity data to synthetic training emission factors, calculate FY2024 and FY2025 Scope 1 and Scope 2 emissions, and compare absolute and revenue-intensity results. You will ask AI to review the workpaper structure and draft observations, not to supply factors.

You'll build

An emissions-workpaper.xlsx or emissions-workpaper.csv set containing calculation rows, annual summaries, intensity metrics, control checks, a variance note and complete factor provenance.

1 Confirm boundaries

2 Join factors

3 Calculate and reconcile

4 Interpret cautiously

Tools: Spreadsheet · approved AI assistant · structured-esg-data.csv · emission-factors-training.csv

Calculate Scope 1, Scope 2 and Intensity Metrics

Test it

The workpaper must reproduce the expected training totals within 0.01 tCO₂e and the displayed intensities within 0.01 tCO₂e/S\$ million. Every calculation row must retain Calculation_ID, activity source, Transformation_ID, factor source HLF-FACTOR-TRAINING, factor ID, units and formula. C05–C09 must be PASS or have a named OPEN owner. The variance note must distinguish absolute and intensity movement and must not state an unsupported cause. prompt-log.csv must contain a completed P-004 row.

Run Materiality, Gap and Visual Analysis

LAB 5 · 50
MIN

You will use synthetic stakeholder and business evidence to score five topics under separate impact and investor lenses, map each topic to the framework register and classify evidence gaps. You will then create a chart and a short summary that avoids causal overreach.

You'll build

A materiality-gap-analysis.xlsx or CSV set with scoring anchors, topic evidence, two separate lens scores, framework-gap status, a labelled chart, reviewer decisions and a follow-up plan.

1 Set scoring anchors

2 Score with evidence

3 Map the gaps

4 Visualise and review

Tools: Spreadsheet · approved AI assistant · stakeholder-impact-notes.md · framework-and-applicability-register.csv

Run Materiality, Gap and Visual Analysis

✓ Test it

MAT-GAP-01 must contain all five topics with Proposed_* and Final_* component scores, formulas based on Final_* scores, source IDs, evidence strength, final reviewer and any adjustment reason. Impact and investor totals must remain separate and range from 0–8. Every gap must map to a named candidate requirement and owner. The chart must encode both lenses visibly, label axes and topics, include the synthetic-source limitation and reconcile to the data table. prompt-log.csv must contain P-005.

Recap — AI for ESG Data Collection and Analysis

- 1 You can now: LO2: structure ESG source data with traceable provenance, units and quality status
- 2 You can now: LO2: calculate and interpret selected emissions and intensity metrics with transparent lineage
- 3 You can now: LO2: perform transparent materiality and gap analysis and communicate results with an honest visual

TOPIC 03

AI for Drafting and Structuring Reports

Evidence-grounded narrative · executive summaries · report architecture · tone and readability · claim validation

Key Concepts — AI for Drafting and Structuring Reports

1

Disclosure logic

State boundary and method before performance, interpretation, actions and limitations.

2

Claim ledger

Link every quantitative and material qualitative claim to evidence and review status.

3

Connected information

Keep governance, strategy, risks, metrics, targets and financial effects coherent.

4

Plain language

Prefer precise verbs, defined terms, short sentences and visible uncertainty.

5

Consistency controls

Lock names, periods, units, totals, targets, baselines and framework references.

6

Fact-check loop

Extract claims, verify sources and calculations, repair or remove unsupported wording.

Draft Narrative Disclosures and Executive Summaries

WHAT

- A disclosure narrative converts verified evidence into a balanced account of context, method, performance, actions and limitations. An executive summary selects only the most decision-relevant messages.

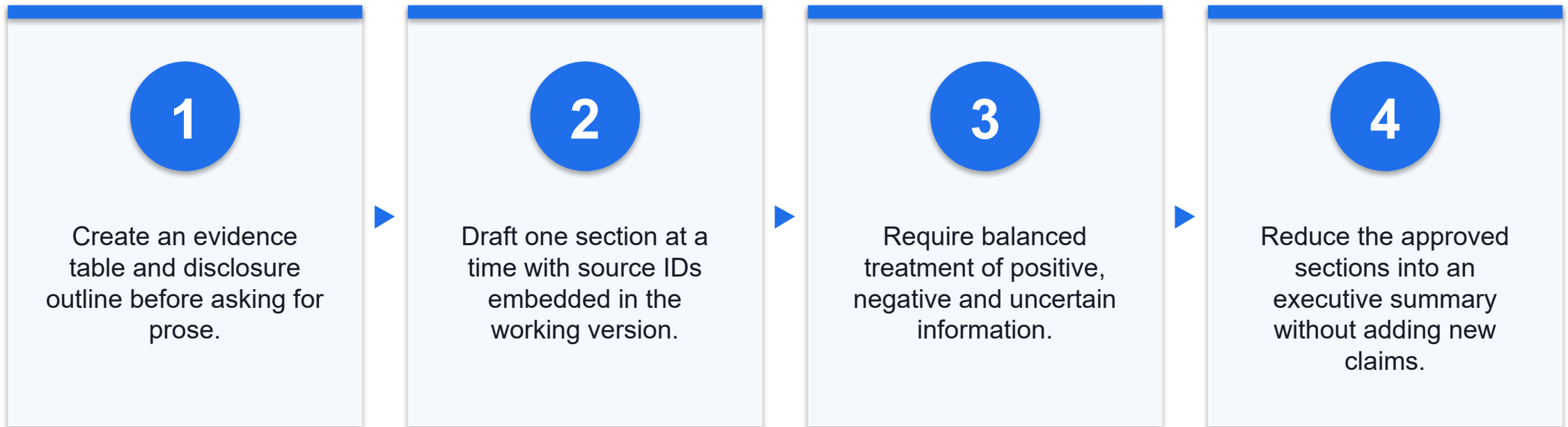
WHY IT MATTERS

- AI is useful for structure and variation, but it tends to smooth over caveats and make ordinary changes sound strategic. A source-led outline prevents language quality from outrunning evidence quality.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Draft Narrative Disclosures and Executive Summaries



In Practice — Draft Narrative Disclosures and Executive Summaries

WORKED EXAMPLE

- The energy section states boundary, method, absolute result, intensity result and limitation.
- A decrease is described as observed; its cause remains unconfirmed pending operations evidence.
- The executive summary reuses approved claims and includes one priority data gap.

DECISION GUIDE

- **Use when**
 - The evidence table is complete enough to support the intended claims.
 - A human editor owns materiality, balance and final wording.
- **Avoid when**
 - The prompt asks for a complete report before framework mapping and data validation.
 - Positive tone is used to remove adverse results, uncertainty or missed targets.

Quality Lens — Draft Narrative Disclosures and Executive Summaries

Failure signal

- The narrative contains strategy claims that do not appear in source records.

Repair move

- Return to the evidence table and delete, qualify or source every unsupported claim.

Quality evidence

- The executive summary contains no claim absent from approved disclosures.

Structure Reports and Sections with AI

WHAT

- Report structure is the intentional hierarchy that connects reporting basis, governance, strategy, material topics, performance, targets, methods and reference indexes.

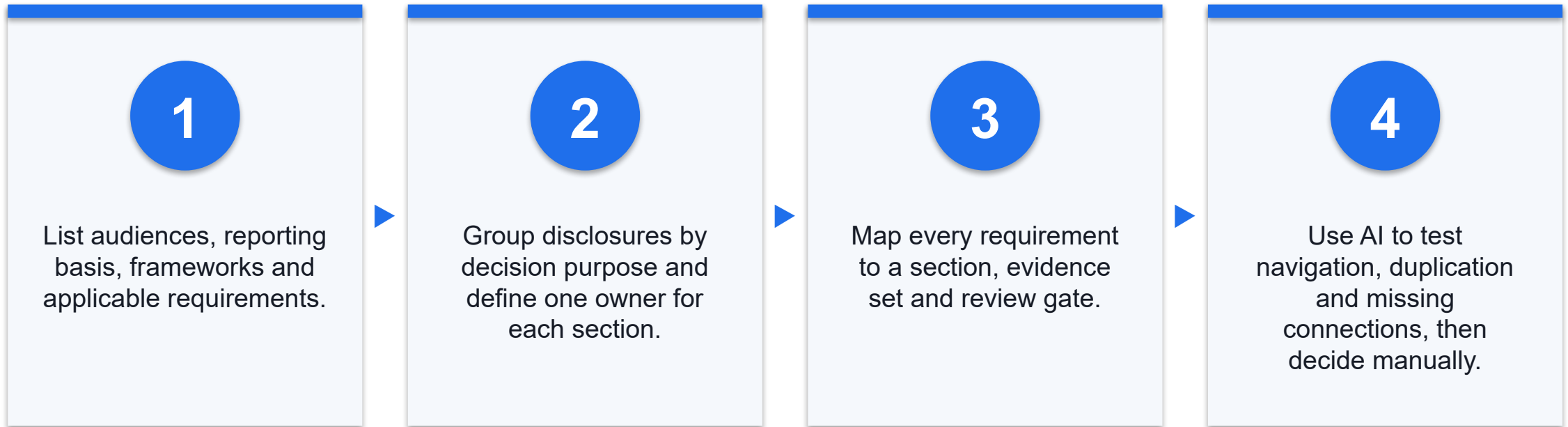
WHY IT MATTERS

- A long generated table of contents can duplicate content or separate metrics from their methods. A requirements-to-section map makes structure serve coverage and reader navigation.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Structure Reports and Sections with AI



In Practice — Structure Reports and Sections with AI

WORKED EXAMPLE

- Governance and strategy appear once, with cross-references from material topic sections.
- Metric tables include basis, boundary and methods immediately before the results.
- The GRI content index and ISSB cross-reference table point to approved sections.

DECISION GUIDE

- **Use when**

- The structure is driven by requirements and reader questions rather than generated headings.
- Cross-references can be maintained without duplicating inconsistent claims.

- **Avoid when**

- The assistant invents a generic report architecture before applicability is established.
- The same target or metric is restated differently in several sections.

Quality Lens — Structure Reports and Sections with AI

Failure signal

- A requirement maps to multiple sections with different owners and values.

Repair move

- Assign a canonical source and one approved statement, then cross-reference it.

Quality evidence

- Each requirement, section, owner and evidence set has a single current status.

Ensure Consistency, Tone and Readability

WHAT

- Consistency control checks facts, terminology, units, periods, targets and narrative stance across the report. Readability makes complex information understandable without removing necessary precision.

WHY IT MATTERS

- Large reports are edited by many people. AI can detect differences, but it can also standardise away important distinctions or introduce a preferred term that changes a defined meaning.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Ensure Consistency, Tone and Readability



In Practice — Ensure Consistency, Tone and Readability

WORKED EXAMPLE

- The style sheet specifies tCO₂e, FY2025 and 'HarbourLight Foods Pte Ltd'.
- A consistency check finds 212.16 tCO₂e in the table but 221 tCO₂e in the summary.
- The editor corrects the summary from the approved calculation, not from the AI suggestion.

DECISION GUIDE

- **Use when**
 - Canonical facts and terminology are available for comparison.
 - The editor can compare original and revised meaning line by line.
- **Avoid when**
 - A 'make this stronger' prompt encourages promotional or absolute claims.
 - Readability edits remove boundaries, assumptions or material caveats.

Quality Lens — Ensure Consistency, Tone and Readability

Failure signal

- The same metric appears with different values, units or periods.

Repair move

- Link every repetition to a canonical source and rerun contradiction checks.

Quality evidence

- All repeated facts reconcile and the plain-language version preserves limitations.

Fact-Check and Validate AI Output

WHAT

- Validation decomposes AI output into checkable claims and tests each one against source evidence, calculations, current official requirements and the intended reporting boundary.

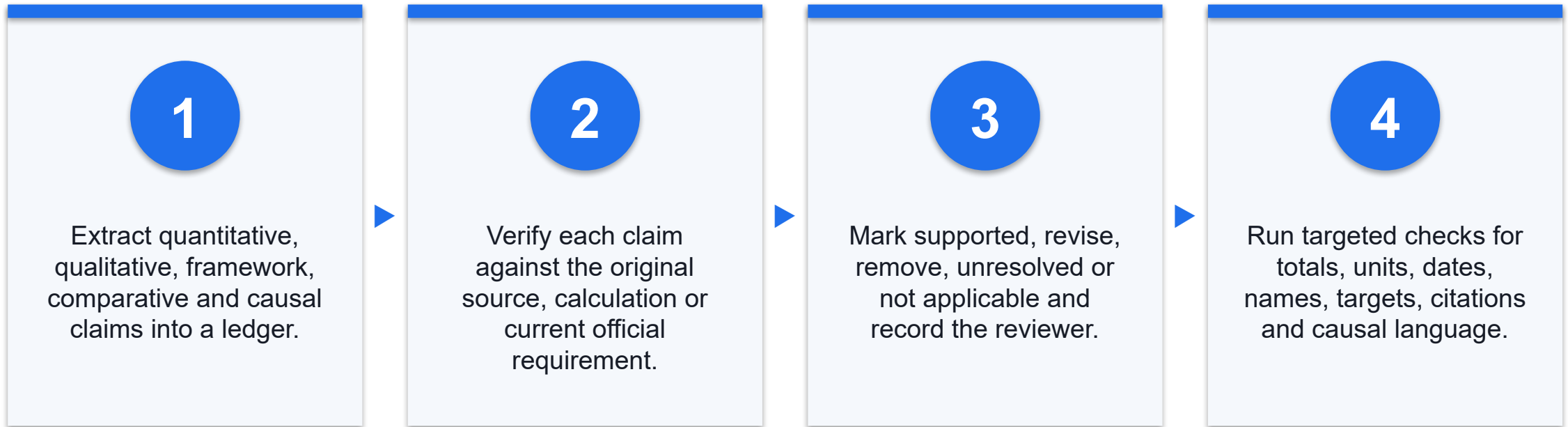
WHY IT MATTERS

- An AI response may contain confabulated facts, inaccurate citations, outdated requirements or unjustified causation. Reviewing overall plausibility is not enough.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Fact-Check and Validate AI Output



In Practice — Fact-Check and Validate AI Output

WORKED EXAMPLE

- Claim C-014 states emissions fell because of equipment upgrades.
- The dataset confirms the fall, but no source record confirms the cause.
- The reviewer changes the wording to an observation and logs the cause as unresolved.

DECISION GUIDE

- **Use when**
 - The output can be decomposed into claims and relevant evidence is accessible.
 - A separate reviewer can challenge high-impact or high-uncertainty items.
- **Avoid when**
 - The same model is treated as the sole judge of its own accuracy.
 - A citation is accepted because its title looks credible without opening the source.

Quality Lens — Fact-Check and Validate AI Output

Failure signal

- The review says 'looks accurate' without claim-level status.

Repair move

- Build a claim ledger and prioritise quantitative, compliance and causal claims.

Quality evidence

- Every retained material claim has a source, reviewer, status and resolution.

Hands-On Labs — AI for Drafting and Structuring Reports

Lab 6

- Draft and Validate an Evidence-Grounded Disclosure

Draft and Validate an Evidence-Grounded Disclosure

LAB 6 · 60
MIN

You will combine the approved reporting basis, emissions workpaper, materiality-gap analysis and a deliberately flawed prior-year disclosure into a controlled FY2025 draft. You will use AI for bounded drafting and critique, then verify every quantitative, causal, framework and target claim.

You'll build

A style-sheet.md, disclosure-draft.md, claim-ledger.csv and executive-summary.md with source-linked claims, consistent units, balanced wording, unresolved limitations and recorded human decisions.

1 Outline from evidence

2 Draft with source tags

3 Build the claim ledger

4 Validate and summarise

Tools: Text editor · spreadsheet · approved AI assistant · disclosure-starters · prior-year-disclosure.md · Lab 4–5 outputs

Draft and Validate an Evidence-Grounded Disclosure

Test it

The final disclosure must be 300–400 words and the executive summary 100–130 words. Both must match the canonical FY2025 Scope 1, Scope 2, total and intensity values from Lab 4. Every retained material claim must have a SUPPORTED or resolved REVISE row in claim-ledger.csv. No UNRESOLVED claim may appear as fact, no unsupported cause or target may remain, and the consistency log must contain no open quantitative conflict. prompt-log.csv must contain completed P-006 review fields and every retained change must be visible in the claim ledger.

Recap — AI for Drafting and Structuring Reports

1

You can now: LO3: draft readable sustainability disclosures and validate every material claim

TOPIC 04

AI for Compliance, Frameworks and Continuous Reporting

Framework crosswalks · current applicability · assurance readiness · governance · recurring and multi-year workflows

Key Concepts — AI for Compliance, Frameworks and Continuous Reporting

1

Applicability register

Record entity scope, jurisdiction, framework basis, effective date and source owner.

2

Requirement crosswalk

Map identifiers to report sections, evidence, owner and status without claiming more than supported.

3

Audit trail

Retain source, transformation, calculation, prompt, review and approval lineage.

4

Segregation of duties

Separate preparer, reviewer and approver for material claims and calculations.

5

Change control

Detect changes in data, factors, methods, requirements and approved narrative.

6

Recurring workflow

Use a calendar, data contracts, controlled templates, quality gates and post-cycle improvement.

Map Content to GRI, SASB, TCFD and ISSB

WHAT

- A framework crosswalk records each requirement or guidance item, why it applies, where it is addressed, which evidence supports it and what status remains.

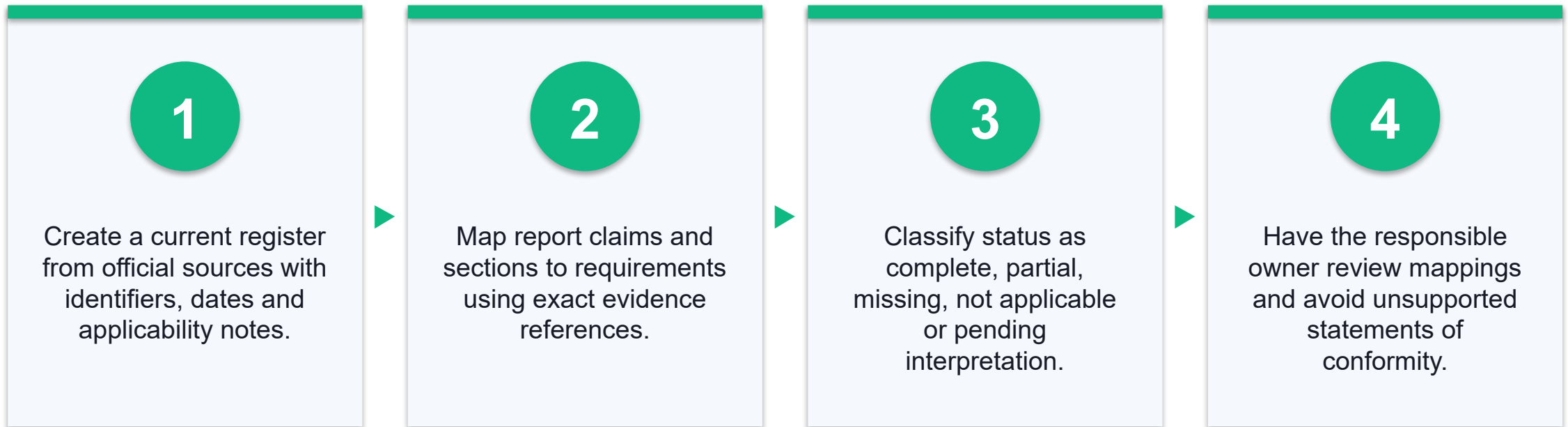
WHY IT MATTERS

- AI can accelerate comparisons, but keyword similarity is not compliance. Definitions, materiality, industry, effective dates and jurisdiction determine whether a mapping is valid.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Map Content to GRI, SASB, TCFD and ISSB



In Practice — Map Content to GRI, SASB, TCFD and ISSB

WORKED EXAMPLE

- GRI 3 supports the documented process for determining impact material topics.
- IFRS S1 and S2 use governance, strategy, risk management, and metrics-and-targets content areas.
- A SASB food-industry topic adds industry detail; TCFD is recorded as a legacy bridge incorporated into IFRS S2.

DECISION GUIDE

- **Use when**
 - Official sources, applicability decisions and requirement IDs are recorded.
 - The crosswalk is a working control with owners and evidence statuses.
- **Avoid when**
 - A generated checklist substitutes for reading the applicable standard or rule.
 - The report claims alignment or conformity when required items remain partial or missing.

Quality Lens — Map Content to GRI, SASB, TCFD and ISSB

Failure signal

- Mappings contain page numbers but no evidence or applicability rationale.

Repair move

- Add requirement ID, lens, official source, owner, evidence and status.

Quality evidence

- A reviewer can trace each mapping and see unresolved items immediately.

Build Assurance-Ready Audit Trails and Governance

WHAT

- Assurance readiness means evidence, methods, controls, roles and changes are documented so an independent reviewer can understand how a reported statement was produced.

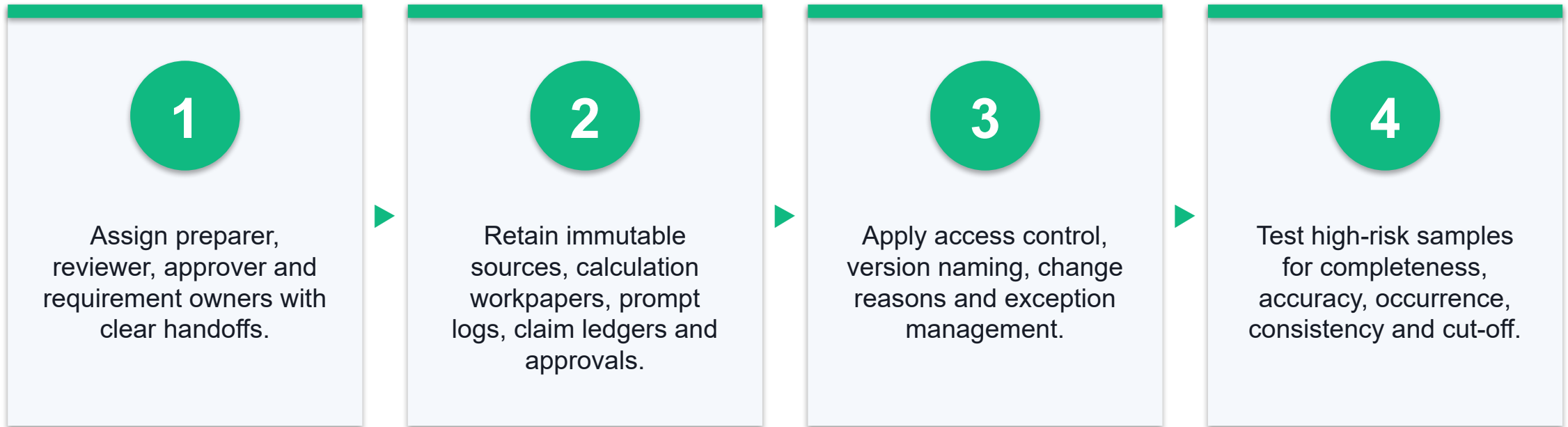
WHY IT MATTERS

- AI introduces another transformation layer. Without logs, version control and approval records, a team may be unable to reproduce a calculation or explain why generated wording was accepted.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Build Assurance-Ready Audit Trails and Governance



In Practice — Build Assurance-Ready Audit Trails and Governance

WORKED EXAMPLE

- Metric M-008 links to invoices, factor EF-03, calculation CALC-02 and reviewer sign-off REV-17.
- Draft D-04 links each retained sentence to claim-ledger rows and an approval record.
- A changed factor triggers recalculation, narrative review and a documented restatement decision.

DECISION GUIDE

- **Use when**
 - Material metrics and claims need reproducible lineage and independent review.
 - The team wants to reduce late-cycle evidence requests and rework.
- **Avoid when**
 - AI logs contain restricted source data that should not be retained in that system.
 - Approval is implied by silence or performed by the same person for every high-risk item.

Quality Lens — Build Assurance-Ready Audit Trails and Governance

Failure signal

- The final report exists but its working papers cannot reproduce key figures.

Repair move

- Reconstruct lineage from source through calculation, draft, review and approval.

Quality evidence

- A reviewer can select a claim and follow its complete evidence chain.

Automate Recurring and Multi-Year Reporting

WHAT

- Recurring reporting uses controlled templates, data contracts, calendars and exception workflows to repeat stable tasks while preserving human review for material judgement and change.

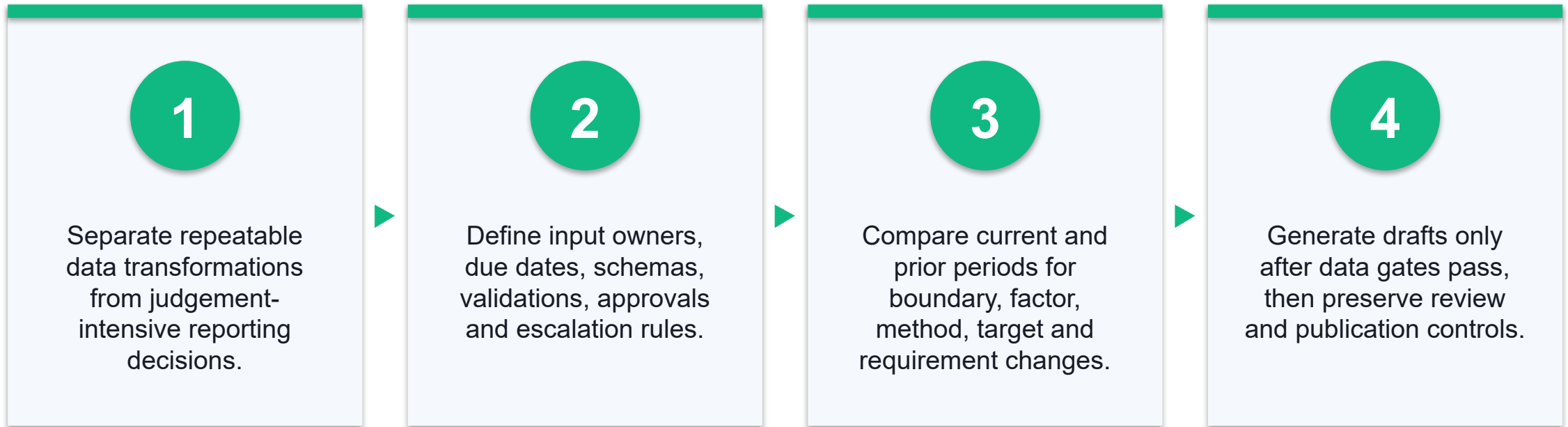
WHY IT MATTERS

- Copying last year's report carries forward stale claims and methods. Uncontrolled automation can reproduce errors faster and conceal changes in boundaries, factors or requirements.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Automate Recurring and Multi-Year Reporting



In Practice — Automate Recurring and Multi-Year Reporting

WORKED EXAMPLE

- Monthly source files load into a locked schema with duplicate and range checks.
- The annual close flags new sites, factor changes and restatement triggers before trend calculations.
- AI drafts variance commentary from approved data, while owners verify explanations and actions.

DECISION GUIDE

- **Use when**

- Inputs are standardised, controls are testable and exceptions have named owners.
- The process keeps human approval for materiality, estimates, causal claims and publication.

- **Avoid when**

- The workflow republishes prior narrative without current evidence.
- Automation is introduced before definitions, ownership and quality gates are stable.

Quality Lens — Automate Recurring and Multi-Year Reporting

Failure signal

- The process is fast but no one owns rejected records or changed requirements.

Repair move

- Add exception queues, owners, service levels and change-control gates.

Quality evidence

- The cycle is repeatable, exceptions are visible and all published changes are approved.

Build an Efficient Sustainability Reporting Workflow

WHAT

- An efficient workflow sequences scoping, evidence intake, calculation, materiality, drafting, framework mapping, review, approval and publication so defects are caught near their source.

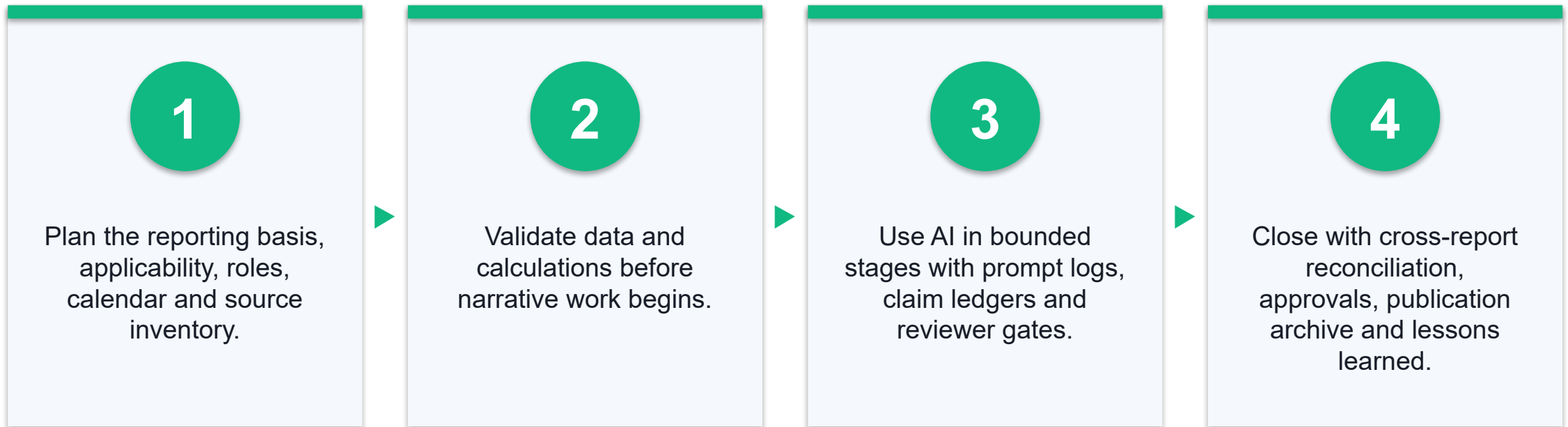
WHY IT MATTERS

- Late reconciliation is expensive. Efficiency comes from clear entry criteria and controls, not from generating more text earlier in the cycle.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Build an Efficient Sustainability Reporting Workflow



In Practice — Build an Efficient Sustainability Reporting Workflow

WORKED EXAMPLE

- HarbourLight's eight-stage board assigns one owner, entry gate and output to each stage.
- A missing energy invoice blocks the metric and narrative but not unrelated governance sections.
- The post-cycle review converts recurring defects into next-year data-contract improvements.

DECISION GUIDE

- **Use when**

- The team needs an end-to-end operating model and clear handoffs.
- Management wants transparent status, exceptions and evidence readiness.

- **Avoid when**

- A single AI prompt is expected to perform data validation, materiality, drafting and sign-off.
- Speed is measured only by draft completion rather than rework, evidence gaps and review findings.

Quality Lens — Build an Efficient Sustainability Reporting Workflow

Failure signal

- Drafting begins while scope, data owners or framework basis remain undecided.

Repair move

- Introduce entry gates and stop affected work until foundational decisions are recorded.

Quality evidence

- Each stage has an owner, input, control, output, status and escalation path.

Hands-On Labs — AI for Compliance, Frameworks and Continuous Reporting

Lab 7

- Build the Framework Crosswalk and Assurance Evidence Index

Lab 8

- Design the Recurring Reporting Workflow and Final Pack

Build the Framework Crosswalk and Assurance Evidence Index

LAB 7 · 55 MIN

You will extend the framework register into a requirement crosswalk, map the Lab 6 disclosure and Lab 4 metrics, and build an evidence index from source through calculation, claim and approval. You will also record current Singapore applicability questions for responsible-owner confirmation.

You'll build

A framework-crosswalk.csv, assurance-evidence-index.csv and control-matrix.csv with requirement IDs, report locations, evidence lineage, owners, statuses, exceptions and reviewer sign-off.

1 Confirm applicability

2 Map requirements

3 Index evidence

4 Test the lineage

Tools: Spreadsheet · browser · approved AI assistant · Lab 2 register · Lab 4 workpaper · Lab 6 claim ledger

Build the Framework Crosswalk and Assurance Evidence Index

Test it

Every ACTIVE crosswalk row must contain an official source, requirement or guidance ID, lens, applicability status, report location, evidence status and owner; every REFERENCE row must retain its current official URL and checked or PENDING CHECK status. No synthetic jurisdiction decision may be stated as confirmed. The evidence index must contain complete reproducible chains for the four metrics and factor limitation. CTL-01–CTL-07 must have preparer, reviewer, evidence and status, with no unowned FAIL or OPEN item. prompt-log.csv must contain completed P-007 review fields.

Design the Recurring Reporting Workflow and Final Pack

LAB 8 · 55 MIN

You will design a reporting calendar and workflow board that reuses controlled structures without copying stale claims. You will specify source-owner data contracts, AI task boundaries, review gates, change triggers and a final pack index that lets another preparer continue the process.

You'll build

A reporting-calendar.csv, workflow-board.csv, data-contracts.md, final-pack-index.md and post-cycle-review.md covering the complete scope-to-publication cycle and its evidence-controlled AI tasks.

1 Define the cycle

2 Set data contracts

3 Place quality gates

4 Package and improve

Tools: Spreadsheet · text editor · approved AI assistant · all Lab 1–7 artifacts

Design the Recurring Reporting Workflow and Final Pack

Test it

The calendar must contain all ten phases with owner, reviewer, entry criteria, output and escalation. data-contracts.md must cover all five evidence classes and versioned corrections. The workflow board must contain at least twelve tasks, allowed AI roles only, named human decisions and reopen triggers. The final pack must index every Lab 1–8 artifact, identify open items and give a reproducible Start Here path. The post-cycle review must define all five metrics with the rate numerator/denominator or count/duration Unit_or_Basis, owner and trigger. prompt-log.csv must contain P-008, and final-pack-index.md must be updated after the post-cycle review so every Lab 1–8 artifact is indexed.

Recap — AI for Compliance, Frameworks and Continuous Reporting

1

You can now: LO4: map approved content to current framework sources and maintain an assurance-ready audit trail

2

You can now: LO4: design a governed recurring workflow for multi-year sustainability reporting



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Ground & Govern

Establish a responsible, evidence-controlled generative AI workflow and distinguish the purposes of GRI, SASB, TCFD and ISSB reporting guidance.

2

Structure & Analyse

Structure ESG source data, calculate and interpret selected emissions and intensity metrics, and perform transparent materiality and gap analysis.

3

Draft & Validate

Draft readable, evidence-grounded sustainability disclosures and executive summaries, then validate claims, consistency and limitations.

4

Map & Operationalise

Map report content to applicable frameworks, maintain an assurance-ready audit trail, and design a governed recurring reporting workflow.

A Practical 30-Day Application Plan

1

Week 1 — confirm scope, owners, approved tools and source inventory.

2

Week 2 — pilot one data-to-metric workpaper with complete lineage.

3

Week 3 — draft one disclosure and close every high-risk claim-ledger item.

4

Week 4 — review the workflow, exceptions, rework and next-cycle controls.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Keep your downloaded C1311 course files and use the lab checkpoints to restart safely.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

C1311 · KEEP THE EVIDENCE VISIBLE

Thank You!

You now have an evidence-controlled workflow for collecting, analysing, drafting, mapping and reviewing sustainability information with accountable human oversight.